

More Gauss's Law — the Three Symmetries

PHYS 230/235 · Wed Sep 23 · Knight 24.4–24.6

Name: _____

Goal. Use Gauss's law to *compute* $\vec{E}$ — not just check it — for the three distributions where symmetry makes it possible: an infinite line, an infinite sheet, and a uniformly charged sphere.

1. Why a symmetry argument is needed

Gauss's law, $\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0}$, is true for *every* closed surface. But it only lets you *solve* for E when you can pull E outside the integral. That needs two things to be true everywhere on the surface you choose:

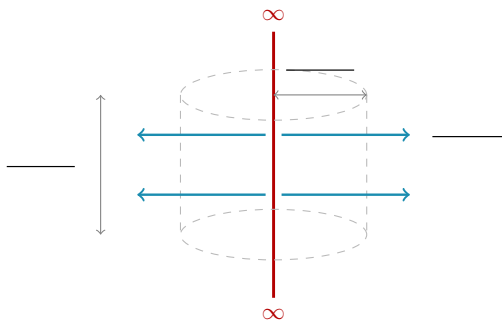
1. E has the same _____ everywhere on it, and
2. $\vec{E}$ is everywhere either _____ or _____ to the surface.

When both hold, the flux integral collapses to a single product: $\Phi = \underline{\hspace{2cm}}$

Check. Does the surface have to match the shape of the *charge*, or the shape of the *field*? Why does a cube fail for a point charge?

2. Infinite line of charge

Linear charge density λ . $\vec{E}$ points radially out from the wire and cannot depend on position along it, so the Gaussian surface is a *coaxial cylinder* of radius r and length L .



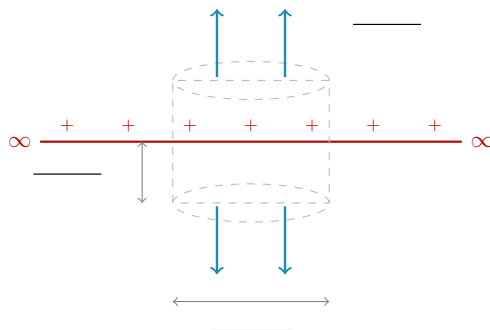
Label the cylinder's radius, its length, and the field arrows.

Which part of the cylinder carries flux (and which carries none)? Area A , then q_{enc} , then solve for E . Show L cancelling.

E for an infinite line

3. Infinite sheet of charge

Surface charge density η . $\vec{E}$ points straight away from the sheet on both sides, so the Gaussian surface is a *pillbox* straddling it, with end caps of area A_{cap} .



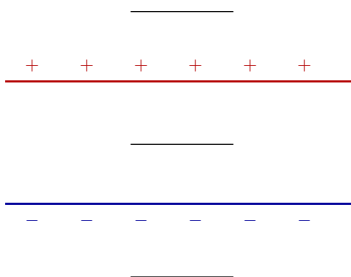
Label the pillbox half-height, the cap area, and the field arrows.

Total area carrying flux, q_{enc} in terms of η and A_{cap} , then E .

E for an infinite sheet

Check. Two surprises. A_{cap} cancelled — why did it *have* to? And the answer contains no r at all. What does that say about walking away from the sheet?

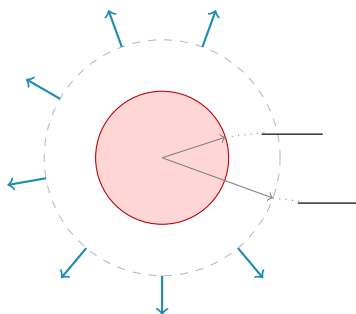
Then: two parallel sheets, one $+\eta$ and one $-\eta$. Superpose the two fields you just found — nothing new to derive.



Write E in each of the three regions: above, between, below.

4. Uniformly charged sphere

Total charge Q spread uniformly through a solid sphere of radius R . There are two regions, and the only difference between them is q_{enc} .



Label the sphere's radius R and the Gaussian radius r .

Outside ($r > R$): the Gaussian sphere encloses _____ of the charge.

q_{enc} , then E . What familiar result does this match?

E outside the sphere

Inside ($r < R$): only the charge within radius r counts. The charge fills the _____, so the fraction enclosed is the ratio of _____.

$q_{\text{enc}} = ?$ then $E = ?$

E inside the sphere

Check. Sketch E against r across both regions. At $r = R$ — jump, kink, or neither?



All three results are in the *Applying Gauss's Law* sim — open its **Info** panel to check yours.